INDEPENDENT PUBLIC-DATA AUDIT / 24 OCTOBER 2026

When can simulation predict real-world robot rankings?

Matched simulation is useful on SIMPLER. This audit finds no improvement from the prespecified task-aware calibrator, and no validated rule for skipping physical tests.

42 paired score cells | **9** task entries | **2** robot embodiments

Held-out evaluation

Every fitted prediction excludes all physical labels from its test task and stays within robot embodiment. B1 uses matched simulation; B2 is affine calibration; C1 adds two frozen task attributes with fixed shrinkage. No hyperparameters are chosen on test outcomes.

Method	Google rank error	Google MAE	WidowX rank error	WidowX MAE
B0 Global simulation	9.7%	20.8 pp	12.5%	9.7 pp
B1 Matched simulation	5.7%	13.9 pp	12.5%	7.7 pp
B2 Affine calibration	5.7%	11.1 pp	12.5%	9.6 pp
C1 Task attributes	5.7%	19.6 pp	12.5%	11.1 pp

Rank error averages tasks within embodiment; only observed real gaps of at least 10 points are eligible. MAE is mean absolute error in percentage points. Five Google and four WidowX tasks underpin these estimates.

What the failures reveal

B1 gets 61/65 Google and 8/9 WidowX comparisons right. These are overlapping pairs, not independent samples. Closing drawers accounts for three material ranking errors. Move Near and Carrot on Plate each add one.

On Move Near, simulation slightly favors RT-1-Begin over Octo-Base (5.0% vs 4.2%); physical success instead favors Octo-Base by 33.3 points. A strong overall correlation can hide consequential local mistakes.

C1 cannot repair individual rankings when its simulation slope is positive. All fitted slopes are positive. Task-only additive effects therefore test calibration, not richer policy-task interactions. C1 also fails to beat B1 on MAE; this is a null for this model, not a general verdict on task-aware methods.

All nine Pearson/MMRV outputs match the pinned authors' functions. Paper/code differences are documented: the paper includes an extra Google policy and two different eggplant outcomes. The repository snapshot remains primary. Unknown trial counts are left blank. LOTO holds out calibrator labels, not the original simulator-design process.

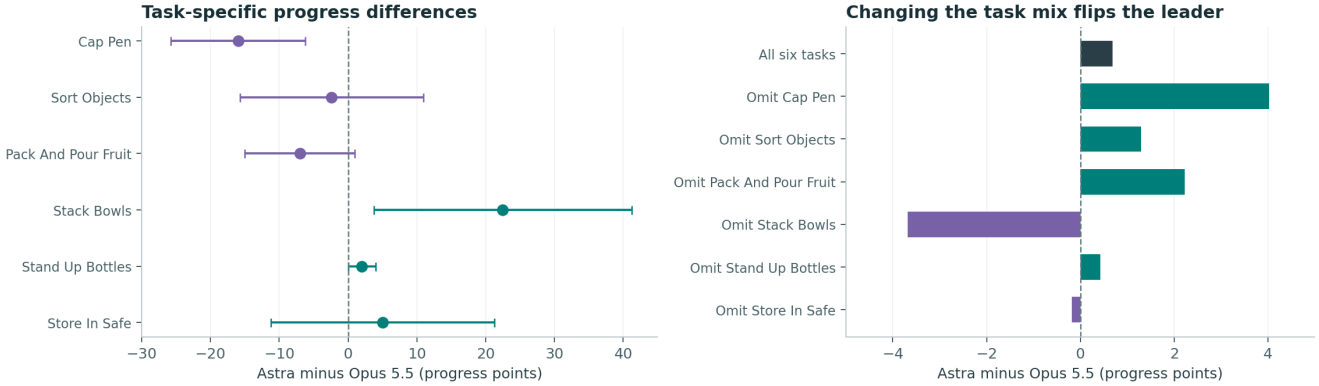
EXTERNAL AUDIT / PHYSICAL STRESS TEST

Task mix can change the winner

Robocurve's 360 trials put Astra at 36.69% progress and Opus 5.5 at 36.01%. Their difference is +0.68 points (conditional 95% bootstrap interval: -4.51 to +5.88).

Robocurve: physical-only stress test

20 trials per model-task. Bars on the left are 95% unpaired bootstrap intervals conditional on these six selected tasks.



Progress is partial-credit performance, not completion. This comparison does not validate simulation.

Astra leads Stack Bowls by 22.5 progress points; Opus 5.5 leads Cap Pen by 16. Omitting Stack Bowls or Store In Safe flips the overall leader. Full completions are only 5/120 for Astra and 1/120 for Opus 5.5. Partial progress and completion remain separate.

RoboDojo: scores found, compatibility still fails

Correction: paper Table 2 provides 180 physical method-task cells (10 methods, 18 tasks; 10 trials per cell). The first website-only audit missed these. But the paper explicitly uses unpaired tasks and separate domain fine-tuning. Shared method names do not identify the same evaluated checkpoint. No matched-task sim-to-real validation is claimed.

As a descriptive cross-task method proxy, overall simulation rankings have 26.8% task-average completion-ranking error on Piper. Piper X has 0% error across only two informative tasks. These are observed gaps, not confident true orderings; one success changes the rate by 10 points.

The next resource is compatible evaluation evidence

Request existing same-checkpoint results on explicitly paired tasks, plus rubric/controller/layout versions. Inspect Cover Blocks and Insert Tubes on ARX X5 first. Request raw repeated-round scores for Store in Safe and Stack Bowls; aggregate stability can conceal task variation.

If a new study becomes necessary, a 12-task, three-family, three-policy pilot with 20 physical trials per cell would total 720 trials. That is a feasibility pilot, not sufficient precision for most 10-point gaps. Freeze a representative task mix and evaluate a trust rule on held-out tasks against B1.

Limits and reproduction

Tasks and checkpoints are few and correlated. Robocurve selected tasks using Astra and uses non-blinded grading and manual resets. Its hardware, agent scaffold and rubrics differ from original RoboDojo. Bootstrap intervals condition on these six tasks and omit rig/time dependence. The exploratory trust score did not yield a defensible deferral rule.

Run **python -m src.reproduce** from the source bundle. Full tables, five figures, source hashes, leakage tests and recommendations are included. No new robot trials, simulation, policy inference or paid APIs were used. Not commissioned or endorsed by Robocurve.

Sources: SIMPLER paper | pinned constants and metrics | RoboDojo paper (Tables 1, 2, 5, 9) | Robocurve report and trial data